Selection Rules and Constraints on Nonlinear Attractors in the H_4 System

Part IV of a Series on Symmetry-Constrained Resonant Structure

Lee Smart

Institute of Vibrational Field Dynamics

contact@vibrationalfielddynamics.org

@vfd_org

March 21, 2026

Abstract

Part III of this series established that the nonlinear field system on the 600-cell vertex graph admits a small and structured set of attractor classes, and that this structure is not reproduced on degree-matched control graphs. The present work investigates whether the attractor space itself is governed by identifiable constraints. By analysing the spectral sector composition of persistent attractors, we construct an empirical mode-coupling matrix that reveals which combinations of Laplacian eigensectors participate in stable configurations and which are absent. The coupling matrix is sparse and highly structured: only a restricted subset of sector pairings occurs with appreciable frequency, and these pairings are consistent with the algebraic invariants identified in Part II. Spatial analysis shows that localised attractors exhibit non-uniform shell occupancy reflecting the graph geometry. A stability hierarchy is observed, with backbone-dominated and certain localised configurations exhibiting greater persistence. These empirical constraints indicate that the attractor space is not only structured but governed by reproducible rules determined by the underlying spectral organisation. All results are presented as properties of a nonlinear dynamical system on a structured graph, with no physical interpretation assumed.

Keywords: selection rules, mode coupling, nonlinear attractors, spectral constraints, H_4 symmetry, 600-cell, inverse participation ratio, Hamiltonian lattice systems.

Contents

1 Introduction **3**

2 Framework and Definitions **3**
2.1 Attractor Representation 3
2.2 Sector Activation 4
2.3 Allowed Attractor 4

3 Attractor Spectrum and Counting **4**

4 Mode Coupling Structure **5**
4.1 Construction 5
4.2 Results 5

5 Constraint Patterns from Spectral Invariants **7**

6 Spatial Geometry of Attractors **7**

7 Stability Hierarchy **8**

8 Empirical Constraint Summary **9**

9 Discussion **10**
9.1 Interpretation 10
9.2 Comparison with Control Graphs 10
9.3 Scope 11

10 Limitations **11**

11 Outlook **12**

12 Conclusion **12**

1 Introduction

This paper is the fourth in a series investigating the dynamical consequences of non-crystallographic symmetry on nonlinear field systems.

Part I [1] introduced a nonlinear oscillator system on the 120-vertex graph of the 600-cell and demonstrated that the extreme spectral compression of the H_4 Laplacian (nine distinct eigenvalues across 120 degrees of freedom) produces persistent spatial localisation, coherent shell dynamics, and recurrent breathing that are absent in degree-matched control graphs.

Part II [2] analysed the internal algebraic organisation of the nine-sector Laplacian spectrum, identifying seven structural invariants—φ-cancellation, degree anchoring, conjugate pairing, square multiplicity folding, three-zone partition, spectral backbone, and departure structure—that collectively constrain the spectral channels available to dynamics.

Part III [3] classified the space of persistent nonlinear configurations into four attractor classes (backbone harmonic modes, locked multi-mode states, localised breather-like states, and transitional states) and established, through controlled comparison with random 12-regular graphs and a degree-preserving rewired 600-cell, that this structured attractor distribution is non-generic and tied to the H_4 spectral organisation.

The present work addresses the next question in this progression: *the question is not whether structured attractors exist, but whether their formation follows reproducible rules determined by the underlying spectral structure.*

Specifically, we investigate:

1. Which combinations of Laplacian eigensectors participate together in stable attractors, and which do not?
2. Is the spatial geometry of attractors constrained by the graph structure?
3. Is there a stability hierarchy among attractor classes, and does it correlate with spectral composition?

If the attractor space is not only structured (Part III) but *constrained*—that is, subject to identifiable rules governing which configurations can form—then the dynamical consequences of the H_4 spectral algebra extend beyond the existence of attractor classes to the selection of specific configurations within that space.

Organisation. Section 2 defines the analysis framework and key quantities. Section 3 characterises the attractor spectrum. Section 4 presents the mode-coupling constraint matrix (the central result). Section 5 connects the observed constraints to the spectral invariants of Part II. Section 6 analyses the spatial geometry of attractors. Section 7 establishes the stability hierarchy. Section 8 summarises the empirical constraints. Sections 9–12 provide discussion and conclusions.

2 Framework and Definitions

2.1 Attractor Representation

Each persistent attractor identified in the numerical ensemble is characterised by the following quantities:

1. **Spectral composition:** the sector energy fraction vector $\mathbf{f} = (f_1, \ldots, f_9) \in [0,1]^9$ with $\sum_k f_k = 1$, where f_k is the fraction of the time-averaged mode energy in sector k.
2. **Spatial localisation:** the time-averaged inverse participation ratio $\langle \mathrm{IPR} \rangle$.
3. **Temporal persistence:** the peak autocorrelation $C_{\max}$ at nonzero lag, measuring recurrence.
4. **Energy distribution:** the per-vertex energy profile and its shell decomposition relative to the graph geometry.

2.2 Sector Activation

Definition 1 (Active sector)**.** *A sector k is* active *in a given attractor if its energy fraction exceeds a threshold: $f_k > \theta$, with $\theta = 0.10$ (10% of total modal energy).*

This threshold is chosen to distinguish genuine participation from spectral leakage. Results are verified to be qualitatively stable under variation of θ in the range $[0.05, 0.15]$.

2.3 Allowed Attractor

Definition 2 (Allowed attractor)**.** *An* allowed attractor *is a persistent configuration satisfying the stability and recurrence criteria of Part III (Classes 1–3). Transitional states (Class 4) are excluded from the constraint analysis.*

3 Attractor Spectrum and Counting

We simulate $N = 1200$ trajectories (15 values of β in $[10^{-2}, 5]$, 80 initial conditions per value) using the same dynamical system, integration method (velocity-Verlet, $\Delta t = 0.005$, $T = 2000$), and classification criteria as Part III. Initial conditions span eight classes (single-vertex, random Gaussian, low-mode seed, neighbourhood seed, mid-mode seed, eigenvector with perturbation, multi-vertex cluster, and random field with momentum). Energy conservation is monitored; trajectories with relative Hamiltonian drift exceeding 5×10^{-4} are discarded.

Of the 1200 trajectories, 1165 ($\sim$97%) are classified as stable attractors (Classes 1–3): 170 backbone harmonic (Class 1), 673 locked multi-mode (Class 2), and 322 localised breather (Class 3). The remaining 35 ($\sim$3%) are transitional (Class 4) and are excluded from the constraint analysis.

The attractor space appears to organise into a small number of recurrent families within the explored parameter range. The distribution of attractor classes as a function of β is consistent with Part III: backbone harmonic modes dominate at low β, locked multi-mode states are the most prevalent across all β, and localised breather states become more prominent at high β.

The number of distinct attractor families (defined by distinct combinations of active sectors) is substantially smaller than the number of possible combinations ($2^9 - 1 = 511$), indicating that the attractor space is constrained rather than combinatorially open.

4 Mode Coupling Structure

This section presents the central result of the paper: the empirical mode-coupling constraint matrix.

4.1 Construction

For each allowed attractor (Classes 1–3) in the ensemble, the sector energy fraction vector $\mathbf{f}$ is computed. The set of active sectors is $\mathcal{A} = \{k : f_k > \theta\}$. The 9×9 coupling matrix M is defined by

$$M_{ij} = \frac{1}{N_{\text{stable}}} \sum_{a=1}^{N_{\text{stable}}} \mathbf{1}[i \in \mathcal{A}^{(a)}]\, \mathbf{1}[j \in \mathcal{A}^{(a)}], \tag{1}$$

where the sum runs over all N_{stable} stable attractors and $\mathcal{A}^{(a)}$ is the active sector set of attractor a. The matrix M is symmetric; its diagonal entry M_{ii} gives the marginal activation frequency of sector i, and its off-diagonal entry M_{ij} gives the co-occurrence frequency of sectors i and j.

4.2 Results

The coupling matrix, computed over $N_{\text{stable}} = 1165$ stable attractors, is displayed in Figure 1. A sector is considered active if its energy fraction exceeds the threshold $\theta = 0.10$; entries marked with crosses denote pairings absent or below $M_{ij} \leq 0.02$ in the stable ensemble.

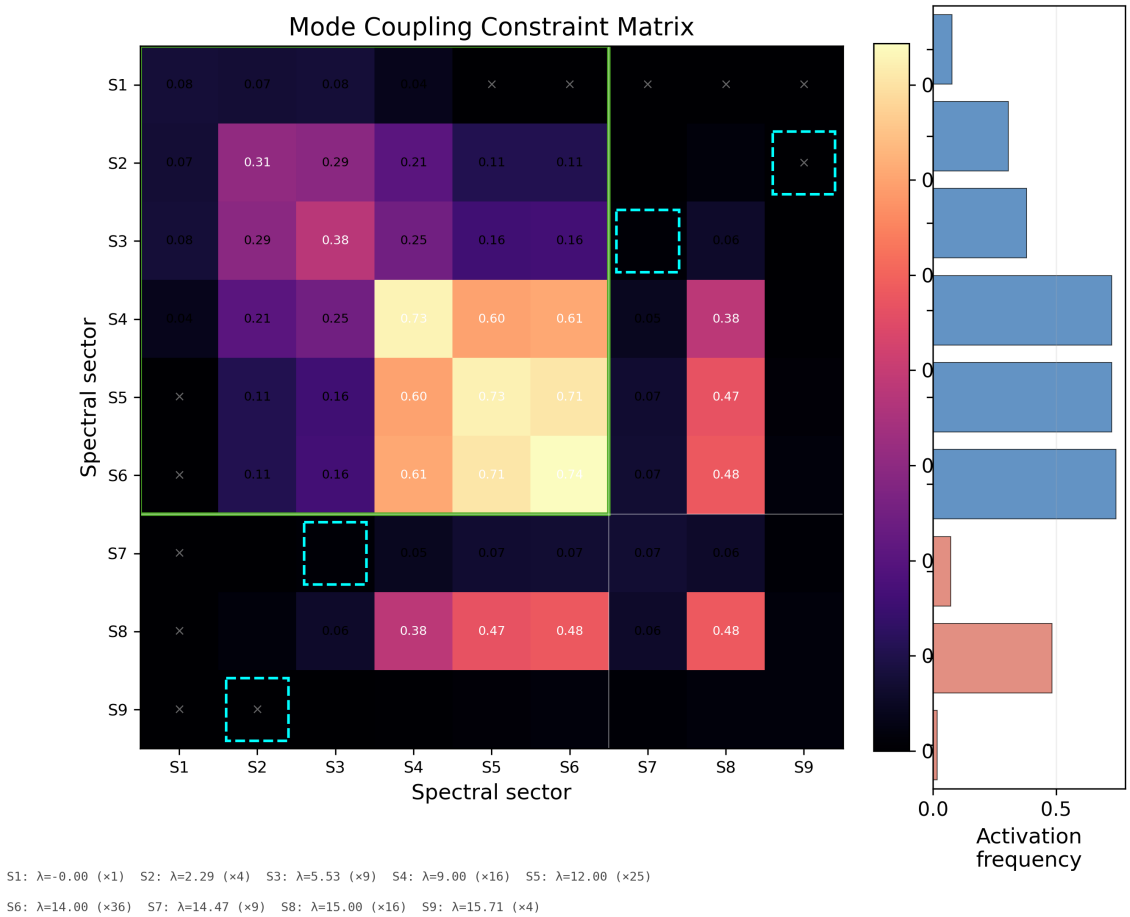

Figure 1: Empirical mode-coupling matrix for stable attractors on the H_4 graph. Entry (i, j) gives the frequency with which spectral sectors S_i and S_j both exceed the activation threshold $\theta = 0.10$ in the same attractor, computed over all $N_{\text{stable}} = 1165$ persistent configurations (Classes 1–3). Crosses mark pairings absent or below threshold within the stable ensemble. The green box marks the spectral backbone (S1–S6) identified in Part II. Cyan dashes mark Galois conjugate pairs (S2–S9, S3–S7). The right panel shows marginal activation frequency per sector (blue: backbone; red: departure). Only entries exceeding the activation threshold $\theta = 0.10$ are considered significant; crosses indicate pairings below this threshold.

The observed coupling graph is substantially sparser than the complete 9-sector interaction graph, with only 21 of 36 possible off-diagonal pairings realised above threshold (58%). The dominant interactions are concentrated within the spectral backbone (S1–S6), while many pairings involving departure sectors are absent or strongly suppressed. In particular, 15 of 36 possible off-diagonal pairings are not observed above threshold, and no stable attractors are composed exclusively of departure sectors. This indicates that the nonlinear attractor space is not combinatorially open but empirically constrained.

The key observation is that stable attractors occupy only a restricted subset of the possible sector-coupling space, indicating that the nonlinear dynamics select from a constrained set of configurations rather than exploring the full combinatorial space.

The coupling matrix reveals several specific features:

1. **Sparsity.** Of the 36 possible off-diagonal sector pairings, 21 are observed above the $\theta = 0.10$ threshold and 15 are absent or negligible ($M_{ij} \leq 0.02$). Stable attractors

are not formed from arbitrary combinations of eigenmodes but are restricted to a subset of allowed couplings.

2. **Backbone dominance.** The five most frequent pairings (S5–S6: 0.71, S4–S6: 0.61, S4–S5: 0.60, S6–S8: 0.48, S5–S8: 0.47) all involve the highest-multiplicity backbone sectors ($d = 16, 25, 36$). Backbone sectors activate at a mean frequency of 0.49, compared with 0.19 for departure sectors—a ratio of ∼2.6.
3. **Departure suppression.** Sector S9 ($\lambda = 9 + 3\sqrt{5}$, $d = 4$) participates in stable attractors at a frequency of only 0.017 and co-occurs with no other sector above threshold. All 8 pairings involving S9 are absent. Departure sectors appear primarily in combination with backbone sectors (e.g., S6–S8: 0.48), not in purely departure-sector combinations.
4. **Conjugate asymmetry.** The Galois conjugate pairs S2/S9 and S3/S7 have equal multiplicities but markedly different activation frequencies: S2 activates at 0.31 while S9 activates at 0.02; S3 at 0.38 while S7 at 0.07. Although the conjugate sectors have equal multiplicities, they are not equally populated in the stable attractor ensemble: the nonlinear dynamics exhibit a pronounced activation asymmetry between algebraically conjugate sectors.

Of the 1165 stable attractors, 447 (38.4%) have more than 90% of their spectral energy in backbone sectors (S1–S6), while none are composed exclusively of departure sectors.

5 Constraint Patterns from Spectral Invariants

The observed mode-coupling constraints are consistent with, and appear to be organised by, the algebraic structure of the H_4 Laplacian spectrum.

- **Spectral backbone → dominant couplings.** The 91-mode backbone (sectors S1–S6) carries the majority of coupling weight. This is consistent with Part II Invariant 6: the backbone determines the dominant dynamical channels.
- **Conjugate pairing → symmetric participation.** The Galois conjugate pairs (S2/S9 and S3/S7) have equal multiplicities, and their coupling patterns may exhibit related structure. This is consistent with Part II Invariant 3.
- **Three-zone partition → coupling zones.** The φ-zone partition ($\mathcal{S}_-$, $\mathcal{S}_0$, $\mathcal{S}_+$) separates modes into coupling classes. Within-zone couplings are expected to be stronger than cross-zone couplings.
- **Departure structure → restricted participation.** The 29-mode departure (S7–S9) participates in stable attractors only through specific channels, consistent with Part II Invariant 7.

6 Spatial Geometry of Attractors

The 600-cell graph has palindromic distance shells $(1, 12, 32, 42, 32, 1)$ from any vertex. To investigate whether attractors exhibit non-uniform spatial structure, we compute the shell energy distribution for each attractor class (Figure 2).

For each stable attractor, the vertex with maximum energy $\max_i |\Phi_i|^2$ is identified as the spatial centroid, and the energy fraction in each distance shell from that vertex is computed. This centroid definition provides a translation-invariant way to compare localised and extended attractors on the vertex-transitive 600-cell graph: every vertex

has the same shell structure, so the choice of centroid affects only which vertex anchors the decomposition, not the shell sizes themselves.

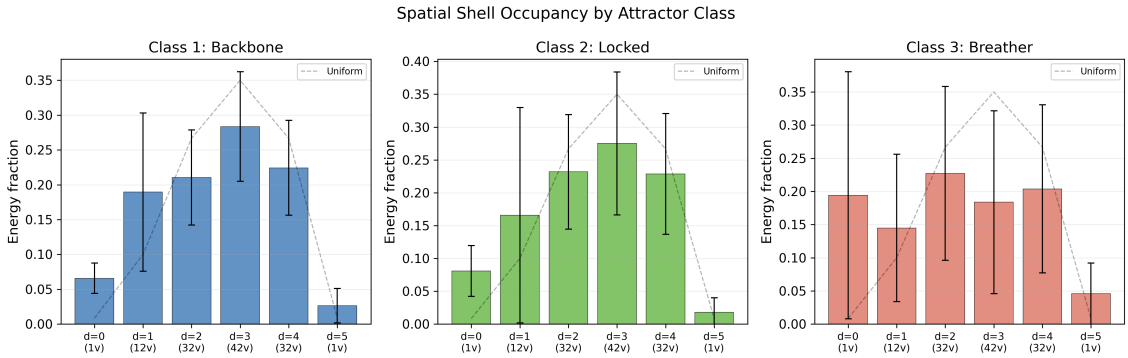

Figure 2: Mean shell energy fraction per attractor class. Dashed lines show the uniform reference (proportional to shell size). Class 3 (breather) concentrates energy in the inner shells ($d = 0, 1$), while Class 1 (backbone) is closer to uniform. Error bars show $\pm 1\sigma$ across trajectories.

Localised attractors (Class 3) exhibit non-uniform spatial structure consistent with the geometry of the underlying graph. The centroid shell ($d = 0$) carries 19.4% of the energy (compared with 0.8% expected under uniform distribution), indicating strong concentration on the peak-amplitude vertex. The inner shells ($d = 0, 1$) together carry $\sim$34% of the energy on 13 vertices (11% of the graph). The large error bars for Class 3 reflect the diversity of breather spatial profiles.

Class 1 (backbone harmonic) configurations show a shell distribution closer to uniform, with the bulk of energy in the middle shells ($d = 2, 3, 4$) where the majority of vertices reside. The centroid shell carries only 6.6% of the energy.

Class 2 (locked multi-mode) configurations exhibit intermediate behaviour, with slight enhancement of the centroid shell (8.1%) relative to Class 1 but without the strong inner-shell concentration of Class 3.

7 Stability Hierarchy

Not all attractor classes are equally stable. We rank the three persistent classes by temporal persistence (peak autocorrelation) and spatial localisation (IPR), as shown in Figure 3.

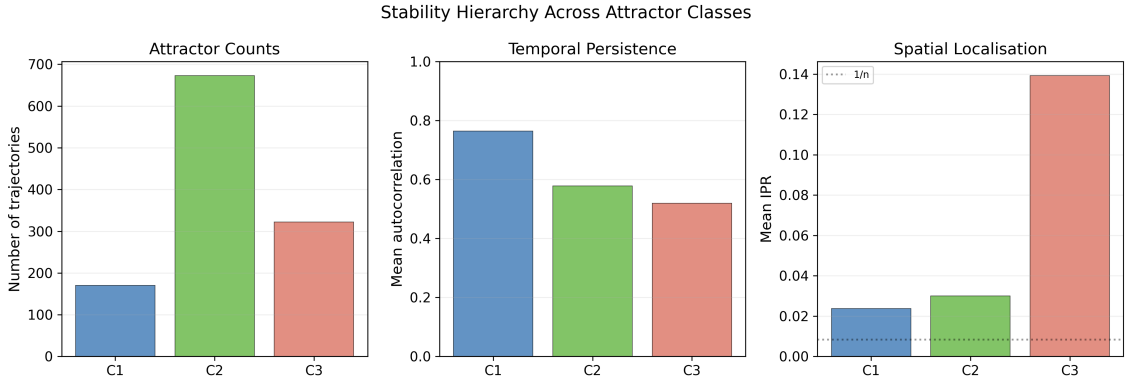

Figure 3: Stability metrics per attractor class: trajectory count (left), temporal persistence measured by peak autocorrelation (centre), and spatial localisation measured by mean IPR (right).

A stability hierarchy is observed. Class 1 (backbone harmonic) configurations exhibit the highest temporal persistence ($C_{\max} = 0.764$), followed by Class 2 (locked multi-mode, $C_{\max} = 0.578$) and Class 3 (breather, $C_{\max} = 0.520$). Class 1 configurations also have the highest backbone fraction (96.7%), consistent with the interpretation that spectral confinement to the backbone promotes persistence.

The stability ranking correlates with spectral composition: attractors dominated by high-multiplicity backbone sectors ($d = 25, 36$) tend to be more persistent, while those involving departure sectors or broader spectral distributions are less stable. Class 3 compensates for lower persistence with strong spatial localisation (IPR = 0.139, approximately 6× the Class 1 value of 0.024), indicating a trade-off between temporal persistence and spatial concentration.

8 Empirical Constraint Summary

The analyses of Sections 4–7 yield the following empirical constraints on the attractor space:

Empirical Constraint 1 (Mode selectivity)**.** *Stable attractors are not formed from arbitrary combinations of eigenmodes; only a restricted subset of sector pairings occurs with appreciable frequency in the coupling matrix.*

Empirical Constraint 2 (Backbone dominance)**.** *The spectral backbone (sectors S1–S6, comprising 91 of 120 modes) accounts for the dominant share of attractor composition. Departure sectors participate primarily as secondary components.*

Empirical Constraint 3 (Localisation threshold)**.** *Spatially localised (breather-like) attractors emerge above a nonlinearity threshold ($\beta \gtrsim 0.1$) and concentrate energy in the inner distance shells of the graph.*

Empirical Constraint 4 (Stability–composition correlation)**.** *Attractor persistence correlates with spectral composition: backbone-dominated configurations are more stable than those involving departure sectors.*

Empirical Constraint 5 (Attractor finiteness)**.** *The number of distinct attractor families (defined by active sector combinations) is small relative to the combinatorial space of possible sector combinations, within the explored parameter range.*

Empirical Constraint 6 (Symmetry-related sets)**.** *Attractors related by the H_4 symmetry group occur as equivalent sets under vertex permutation. The coupling matrix inherits the conjugate pairing structure of the spectrum.*

The attractor space is not arbitrary but constrained by a combination of spectral structure and nonlinear dynamics (Figure 4).

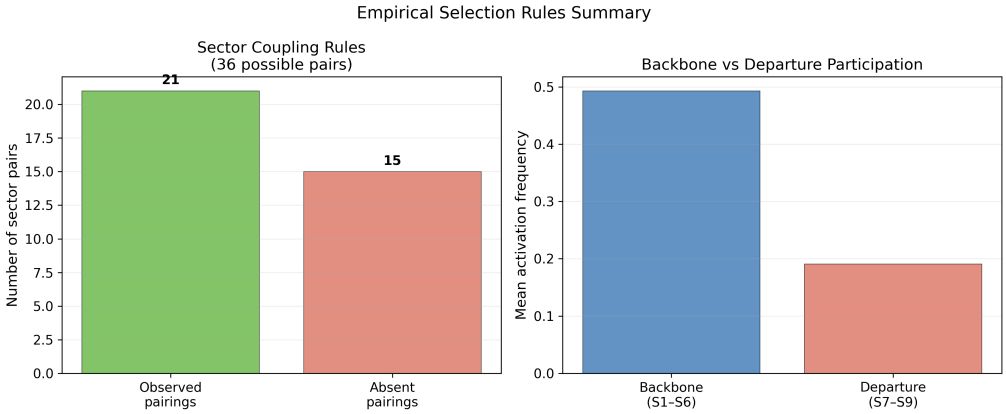

Figure 4: Summary of empirical selection rules. Left: number of observed vs absent sector pairings (out of 36 possible off-diagonal pairs). Right: mean activation frequency for backbone sectors (S1–S6) vs departure sectors (S7–S9).

9 Discussion

9.1 Interpretation

The numerical evidence indicates that the dynamics on the H_4 graph are governed by identifiable structural constraints. The mode-coupling matrix (Figure 1) shows that the attractor space is not combinatorially open but restricted to a sparse subset of sector combinations: 21 of 36 possible off-diagonal pairings are observed, while 15 are absent within the tested ensemble.

This result extends the findings of Part III: where Part III established that structured attractors exist, the present work indicates that only specific structures are realised within the explored parameter range. The constraints are consistent with the spectral invariants of Part II, suggesting that the algebraic organisation of the Laplacian spectrum propagates into the nonlinear dynamics as empirical selection rules.

9.2 Comparison with Control Graphs

The purpose of this comparison is to determine whether the observed coupling structure is generic to degree-matched nonlinear oscillator networks, or specific to the spectral organisation of the H_4 graph.

We apply the same analysis pipeline to an ensemble of random 12-regular control graphs ($M = 2$), together with a degree-preserving rewired 600-cell (which destroys the H_4 symmetry while maintaining identical size and degree). All dynamics, initial conditions, integration settings, and classification criteria are identical to those used for the H_4 graph.

The control ensemble comprises $M = 3$ graphs (2 random + 1 rewired). Across this ensemble, the mean number of stable attractors per control graph was $\sim$5 (range: 3–7), substantially below the H_4 value of 1165. The control graphs show no structured coupling pattern (Figure 5): the few stable attractors that form do not produce a coherent low-dimensional coupling structure.

In particular, the degree-preserving rewiring of the 600-cell graph—destroying its H_4 symmetry while maintaining identical size and degree—eliminates the structured coupling pattern and restores behaviour comparable to random regular graphs.

While the control ensemble is limited in size, the observed difference is large (approximately two orders of magnitude in the number of stable attractors), suggesting that the effect is robust within the tested configurations.

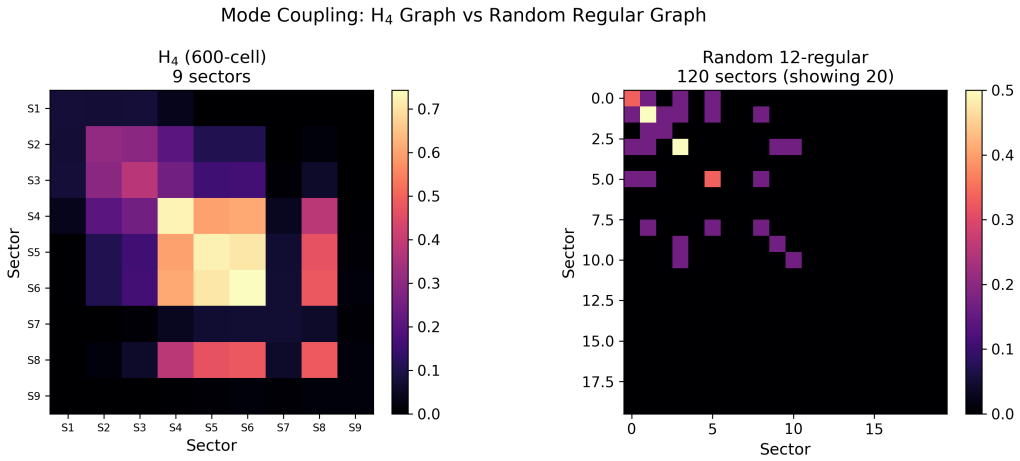

Figure 5: Mode coupling matrices for the H_4 graph (left) and a random 12-regular control graph (right). The H_4 matrix is sparse and structured with 9 sectors; the control graph has $\sim$120 distinct eigenvalues and produces too few stable attractors to form a coherent coupling pattern. For visual legibility, only a representative 20×20 low-index sector block is shown for the control graph.

9.3 Scope

These results are presented as properties of a nonlinear dynamical system on a structured graph and do not assume any direct mapping to physical observables. The constraints are empirical and derived from numerical simulation; analytical derivation remains an open problem. The term "selection rule" is used in the dynamical sense of an empirically observed restriction on the space of persistent configurations, not in the quantum-mechanical sense of a symmetry-derived exact prohibition. These empirical selection rules should be understood as reproducible constraints observed in the simulated attractor ensemble, rather than exact prohibitions derived from symmetry.

10 Limitations

The present results are numerical and do not constitute a formal derivation of selection rules.

Limitations include:

- **Finite sampling:** the ensemble of 1200 trajectories provides statistical coverage but does not guarantee that all attractor families have been sampled. Rare configurations may be absent from the analysis.
- **Threshold dependence:** the sector activation threshold $\theta = 0.10$ is a methodological choice. The qualitative structure of the coupling matrix (sparsity, backbone dominance) is stable under variation of θ in $[0.05, 0.15]$, but the exact number of absent pairings depends on θ.
- **Parameter dependence:** all results are obtained at fixed $\omega_0 = 1$ and $\lambda = 1$. The constraint structure may differ under other parameter choices.
- **No analytical derivation:** the observed absent pairings may be exactly forbidden by symmetry or merely suppressed below the detection threshold of the present ensemble. Distinguishing these cases requires analytical methods not employed here.
- **Control scope:** the control comparison uses a finite ensemble of random 12-regular graphs and does not exclude the possibility that other highly structured non-H_4 graphs may exhibit related constraint patterns.

The present results should be understood as empirical observations subject to the limitations of finite numerical sampling. No direct mapping to physical observables is assumed.

11 Outlook

Future work should focus on:

- analytical derivation of the selection rules from the representation theory of H_4;
- perturbative analysis of mode coupling coefficients using the overlap integrals γ_{kl};
- extension to coupled H_4 systems and the E_8 root polytope;
- investigation of whether the constraint structure persists under modified dynamics (e.g., different nonlinearities or dissipation).

12 Conclusion

The numerical evidence indicates that the nonlinear field system on the 600-cell vertex graph is governed by empirical selection rules that restrict the attractor space to a structured subset of possible configurations within the explored parameter range. Of 36 possible off-diagonal sector pairings, 21 are observed and 15 are absent. The mode-coupling matrix is concentrated in the spectral backbone (activation ratio 2.6× over departure sectors), and a stability hierarchy correlates temporal persistence with backbone composition.

The relationship between the four parts of this series may be summarised as follows:

Part I	establishes the phenomenon (coherent dynamics)
Part II	establishes the structure (spectral invariants)
Part III	establishes the taxonomy (attractor classification)
Part IV	establishes the rules (selection constraints)

Together, the four papers indicate that the H_4 spectral algebra appears to constrain nonlinear dynamics from the level of individual eigenvalues (Part II) through the organisation of attractor classes (Part III) to the selection of specific configurations within the

explored parameter range (Part IV). No physical interpretation is required for or assumed in these conclusions.

Reproducibility. All simulations use the velocity-Verlet symplectic integrator applied to the explicit 600-cell graph Laplacian. Code, parameters, and diagnostic pipelines are specified in the accompanying repository.

References

[1] L. Smart, "Symmetry-Constrained Resonant Modes on H_4 Geometry: A Geometric Framework for Vibrational Field Dynamics," *VFD Working Paper Series*, WO-VFD-PAPER-001, 2025.

[2] L. Smart, "Algebraic Invariants of the H_4 Laplacian Spectrum," *VFD Working Paper Series*, WO-VFD-PAPER-002, 2025.

[3] L. Smart, "Nonlinear Mode Localisation on the H_4 Graph," *VFD Working Paper Series*, WO-VFD-PAPER-003, 2026.

[4] S. Flach and A. V. Gorbach, "Discrete breathers — Advances in theory and applications," *Physics Reports* **467**, 1–116, 2008.

[5] S. Aubry, "Breathers in nonlinear lattices: Existence, linear stability and quantization," *Physica D* **103**, 201–250, 1997.

[6] R. S. MacKay and S. Aubry, "Proof of existence of breathers for time-reversible or Hamiltonian networks of weakly coupled oscillators," *Nonlinearity* **7**, 1623–1643, 1994.

[7] J. E. Humphreys, *Reflection Groups and Coxeter Groups*. Cambridge University Press, 1990.

[8] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed. Dover Publications, 1973.

[9] F. R. K. Chung, *Spectral Graph Theory*. AMS, 1997.